One-page concept summary

Synaptic plasticity as short-term memory — DataForge 2026

Central claim

A recurrent model can store sequential associations in a fixed-shape synaptic state using local Hebbian outer-product writes, avoiding a growing token-by-token KV buffer. The compression is not free: dense associative memory develops cross-talk, and recall degrades as stored associations approach the effective representational rank.

Mechanism

The memory state is updated as W(t)=lambda W(t-1)+eta(V(t) x K(t)^T). The state shape is fixed, so storage does not grow one slot per token. A cue reads from the same state. The desired target appears alongside interference from other stored associations.

Why it matters now

This reframes a familiar inference problem as a memory-location problem: instead of preserving the entire token history, the model can maintain an evolving internal state. BDH makes that idea concrete by using synaptic plasticity as working memory during inference.

BDH connection

The Dragon Hatchling paper describes working memory as relying on synaptic plasticity with Hebbian learning and reports sparse, positive activation patterns. This project uses those published ideas as the basis for an explainer, not as a claim of reproducing the proprietary training pipeline.

Evidence and limitation

Our interactive toy is intentionally transparent and synthetic. It demonstrates the interference mechanism. BDH benchmark values in the presentation are labeled as reported/precomputed evidence. The open-source Pathway repository states that the public code does not reproduce the 97.4% Sudoku Extreme result out of the box. Remaining questions include fast-to-slow consolidation, sparse-update hardware efficiency, and the true capacity of structured synaptic states.

Comparison

Paradigm	State growth	Key trade-off
Transformer	O(T)	Memory + bandwidth grow with context
Linear attention	O(1)	Finite-state interference
Fast weights	O(1)	Dense cross-talk
BDH	O(1) synaptic state	Sparse graph updates + hardware alignment

Primary literature (2022–2025): Tyulmankov et al. (2022); Shervani-Tabar & Rosenbaum (2023); Agnes & Vogels (2024); Kosowski et al. (2025), arXiv:2509.26507.